

# PROS6-A670-OSB

Serie PROS · R513A · senza serbatoio / without tank



PROS6-A670-OSB

RESA FRIGORIFERA · COOLING CAPACITY

**440** kW

EER · 12/7 °C · AIR 35 °C

**2,96**

RESA NOMINALE · 15 °C · AIR 25 °C

**670** kW

PORTATA ACQUA · WATER FLOW

**75,8** m<sup>3</sup>/h

|                            |                 |                                        |                       |
|----------------------------|-----------------|----------------------------------------|-----------------------|
| Compressori / Compressors  | 2 × vite        | L × P × H (mm)                         | 5.890 × 2.197 × 2.402 |
| Circuiti / Circuits        | 2               | Serbatoio / Tank                       | —                     |
| Ventilatori / Fans         | 10              | Taglia / Size                          | 670                   |
| Refrigerante / Refrigerant | R513A · GWP 631 | Campo aria esterna / Ambient air range | -5 / +50 °C           |

## Resa in sintesi / Capacity at a glance <sup>kW</sup>

| ARIA ESTERNA / AMBIENT AIR                  | 25 °C | 30 °C | 35 °C | 40 °C | 45 °C | 50 °C |
|---------------------------------------------|-------|-------|-------|-------|-------|-------|
| Acqua in uscita 7 °C / Leaving water 7 °C   | 491   | 466   | 440   | 414   | 387   | 360   |
| Acqua in uscita 15 °C / Leaving water 15 °C | 670   | 637   | 604   | 570   | 535   | 499   |

Refrigeratori centralizzati condensati ad aria con compressori a vite, per le grandi potenze industriali e commerciali, con parzializzazione della potenza 25-50-75-100 %.

Centralised air-cooled chillers with screw compressors for large industrial and commercial loads, with 25-50-75-100 % capacity steps.

# Dati tecnici

## Technical data

PROS6 - A670 - OSB

**Prestazioni nominali — acqua in uscita 15 °C, aria 25 °C / Nominal performance — 15 °C leaving water, 25 °C air**

|                                     |                          |
|-------------------------------------|--------------------------|
| Resa frigorifera / Cooling capacity | 669,61 kW                |
| Resa frigorifera / Cooling capacity | 575.865 kcal/h           |
| Potenza assorbita / Power input     | 126,57 kW                |
| EER / EER                           | 5,29                     |
| Portata acqua / Water flow          | 115,17 m <sup>3</sup> /h |

**Condizioni standard — acqua 12/7 °C, aria 35 °C / Standard conditions — 12/7 °C water, 35 °C air**

|                                     |                         |
|-------------------------------------|-------------------------|
| Resa frigorifera / Cooling capacity | 440,50 kW               |
| Potenza assorbita / Power input     | 148,70 kW               |
| EER / EER                           | 2,96                    |
| ESEER / ESEER                       | 3,94                    |
| Portata acqua / Water flow          | 75,77 m <sup>3</sup> /h |

### Componenti / Components

|                                             |                      |
|---------------------------------------------|----------------------|
| Compressori / Compressors                   | 2 × Vite             |
| Circuiti frigoriferi / Refrigerant circuits | 2                    |
| Ventilatori / Fans                          | 10                   |
| Condensatore / Condenser                    | a microcanali        |
| Evaporatore / Evaporator                    | a fascio tubiero     |
| Refrigerante / Refrigerant                  | R513A • GWP 631 • A1 |

### Installazione / Installation

|                                                                                           |                           |
|-------------------------------------------------------------------------------------------|---------------------------|
| Dimensioni L × P × H / Dimensions L × D × H                                               | 5.890 × 2.197 × 2.402 mm  |
| Campo aria esterna / Ambient air range                                                    | -5 / +50 °C               |
| Dati elettrici, carica gas, peso, attacchi / Electrical data, charge, weight, connections | su richiesta / on request |

### Dotazioni di serie / Standard equipment

- Valvole di intercettazione  
Shut-off valves
- Griglia di protezione condensatore  
Condenser protection grille
- Valvola di espansione elettronica  
Electronic expansion valve
- Condensatore a microcanali  
Micro-channel condenser
- Evaporatore a fascio tubiero  
Shell & tube evaporator
- Display 4"/7"  
4"/7" display
- Pressostati e manometri  
Pressure switches and gauges
- Pressostato differenziale acqua  
Water differential pressure switch
- Presa di servizio 220 V e luce quadro elettrico  
220 V service socket and electrical panel light

### Con glicole / With glycol

| MISCELA  | COEFF.  | 12/7 • 35 °C |
|----------|---------|--------------|
| MEG 20 % | × 0,980 | 431,7 kW     |
| MEG 30 % | × 0,950 | 418,5 kW     |
| MEG 40 % | × 0,920 | 405,3 kW     |
| MPG 20 % | × 0,975 | 429,5 kW     |
| MPG 30 % | × 0,960 | 422,9 kW     |
| MPG 40 % | × 0,950 | 418,5 kW     |

# Prestazioni Performance

PROS6-A670-OSB · acqua pura,  $\Delta T$  5 K · pure water, 5 K  $\Delta T$  ·  $\square$  12/7 °C · 35 °C

## Resa frigorifera / Cooling capacity (kW)

| LWT ↓ · air → | 15  | 20  | 25  | 30  | 35  | 40  | 45  | 50  |
|---------------|-----|-----|-----|-----|-----|-----|-----|-----|
| 1             | 422 | 403 | 383 | 363 | 342 | 321 | 299 | 277 |
| 2             | 440 | 420 | 400 | 379 | 357 | 335 | 313 | 290 |
| 3             | 458 | 438 | 417 | 395 | 373 | 350 | 327 | 303 |
| 4             | 478 | 456 | 435 | 412 | 389 | 366 | 341 | 317 |
| 5             | 497 | 475 | 453 | 429 | 406 | 381 | 356 | 331 |
| 6             | 518 | 495 | 472 | 448 | 423 | 398 | 372 | 345 |
| 7             | 538 | 515 | 491 | 466 | 440 | 414 | 387 | 360 |
| 8             | 560 | 536 | 511 | 486 | 459 | 432 | 404 | 376 |
| 9             | 582 | 558 | 532 | 505 | 478 | 450 | 421 | 392 |
| 10            | 606 | 580 | 553 | 526 | 497 | 468 | 439 | 408 |
| 11            | 629 | 603 | 575 | 547 | 517 | 487 | 456 | 425 |
| 12            | 654 | 627 | 598 | 569 | 538 | 507 | 475 | 443 |
| 13            | 679 | 650 | 621 | 591 | 559 | 527 | 494 | 461 |
| 14            | 705 | 676 | 645 | 614 | 582 | 548 | 514 | 480 |
| 15            | 732 | 701 | 670 | 637 | 604 | 570 | 535 | 499 |
| 16            | 759 | 728 | 695 | 662 | 627 | 592 | 556 | 519 |
| 17            | 787 | 755 | 721 | 687 | 651 | 615 | 577 | 539 |

## Potenza assorbita / Power input (kW)

| LWT ↓ · air → | 15  | 20  | 25  | 30  | 35  | 40  | 45  | 50  |
|---------------|-----|-----|-----|-----|-----|-----|-----|-----|
| 1             | 101 | 111 | 121 | 133 | 146 | 160 | 176 | 194 |
| 2             | 102 | 111 | 122 | 134 | 147 | 161 | 177 | 194 |
| 3             | 102 | 112 | 122 | 134 | 147 | 161 | 177 | 195 |
| 4             | 102 | 112 | 123 | 134 | 147 | 162 | 178 | 195 |
| 5             | 103 | 112 | 123 | 135 | 148 | 162 | 178 | 196 |
| 6             | 103 | 113 | 123 | 135 | 148 | 163 | 179 | 197 |
| 7             | 103 | 113 | 124 | 136 | 149 | 163 | 179 | 197 |
| 8             | 104 | 113 | 124 | 136 | 149 | 164 | 180 | 198 |
| 9             | 104 | 114 | 124 | 136 | 149 | 164 | 180 | 198 |
| 10            | 104 | 114 | 125 | 137 | 150 | 165 | 181 | 199 |
| 11            | 105 | 114 | 125 | 137 | 150 | 165 | 181 | 199 |
| 12            | 105 | 115 | 125 | 137 | 151 | 165 | 182 | 200 |
| 13            | 106 | 115 | 126 | 138 | 151 | 166 | 182 | 200 |
| 14            | 106 | 116 | 126 | 138 | 151 | 166 | 183 | 201 |
| 15            | 107 | 116 | 127 | 138 | 152 | 166 | 183 | 201 |
| 16            | 107 | 116 | 127 | 139 | 152 | 167 | 183 | 201 |
| 17            | 108 | 117 | 127 | 139 | 152 | 167 | 184 | 202 |

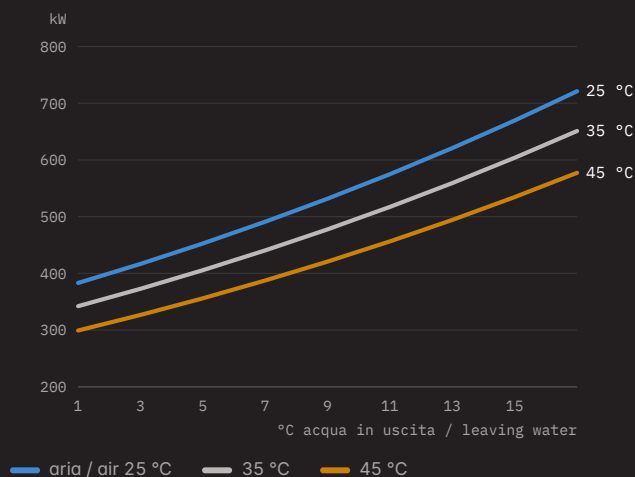
## EER / EER (kW/kW)

| LWT ↓ · air → | 15   | 20   | 25   | 30   | 35   | 40   | 45   | 50   |
|---------------|------|------|------|------|------|------|------|------|
| 1             | 4,17 | 3,63 | 3,15 | 2,72 | 2,34 | 2,00 | 1,70 | 1,43 |
| 2             | 4,33 | 3,78 | 3,28 | 2,84 | 2,44 | 2,08 | 1,77 | 1,49 |
| 3             | 4,49 | 3,92 | 3,41 | 2,95 | 2,54 | 2,17 | 1,84 | 1,56 |
| 4             | 4,67 | 4,08 | 3,55 | 3,07 | 2,64 | 2,26 | 1,92 | 1,62 |
| 5             | 4,84 | 4,23 | 3,68 | 3,19 | 2,74 | 2,35 | 2,00 | 1,69 |
| 6             | 5,03 | 4,40 | 3,83 | 3,31 | 2,85 | 2,44 | 2,08 | 1,76 |
| 7             | 5,21 | 4,56 | 3,97 | 3,44 | 2,96 | 2,54 | 2,16 | 1,83 |
| 8             | 5,40 | 4,73 | 4,13 | 3,57 | 3,08 | 2,64 | 2,25 | 1,90 |
| 9             | 5,60 | 4,91 | 4,28 | 3,71 | 3,20 | 2,74 | 2,33 | 1,97 |
| 10            | 5,80 | 5,09 | 4,44 | 3,85 | 3,32 | 2,85 | 2,43 | 2,05 |
| 11            | 6,00 | 5,27 | 4,60 | 3,99 | 3,44 | 2,95 | 2,52 | 2,13 |
| 12            | 6,21 | 5,46 | 4,77 | 4,14 | 3,57 | 3,07 | 2,62 | 2,22 |
| 13            | 6,43 | 5,65 | 4,94 | 4,29 | 3,70 | 3,18 | 2,71 | 2,30 |
| 14            | 6,65 | 5,85 | 5,11 | 4,45 | 3,84 | 3,30 | 2,82 | 2,39 |
| 15            | 6,86 | 6,05 | 5,29 | 4,61 | 3,98 | 3,42 | 2,92 | 2,48 |
| 16            | 7,09 | 6,25 | 5,48 | 4,77 | 4,13 | 3,55 | 3,03 | 2,58 |
| 17            | 7,32 | 6,46 | 5,66 | 4,93 | 4,27 | 3,68 | 3,14 | 2,67 |

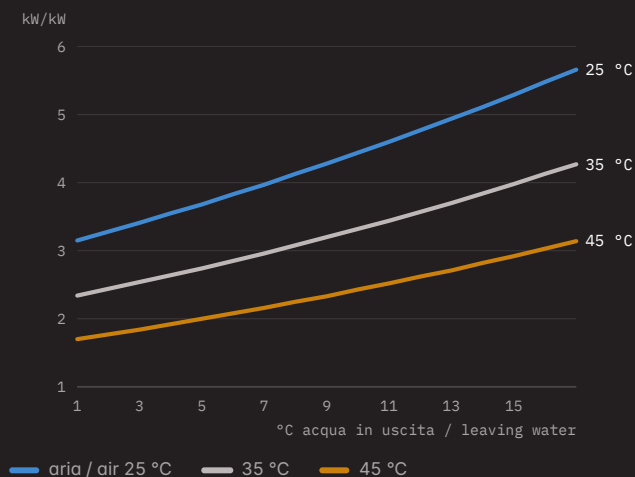
# Curve e dimensioni

## Curves and dimensions

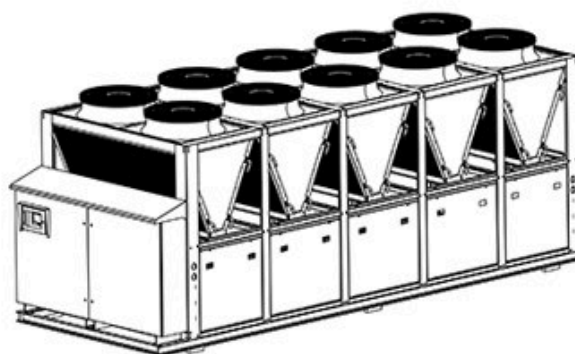
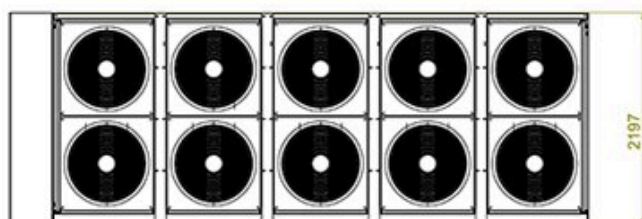
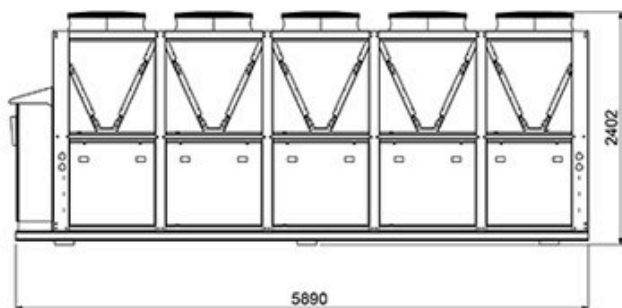
### Resa frigorifera / Cooling capacity



### EER / EER



### Dimensioni / Dimensions · L × P × H 5.890 × 2.197 × 2.402 mm



Disegno indicativo, quote in mm (±10 mm). Dimensioni da confermare in fase d'ordine. / Indicative drawing, dimensions in mm (±10 mm), to be confirmed at order stage.

# Optional — 1/3 Options

PROS6 - A670 - OSB · 75 optional disponibili per questa configurazione · 75 options available for this configuration

## Pompe / Pumps

|                                                                                                    |
|----------------------------------------------------------------------------------------------------|
| Pompa 80 m³/h – prevalenza 25 m<br>Pump 80 m³/h – head 25 m                                        |
| Pompa 80 m³/h – prevalenza 40 m<br>Pump 80 m³/h – head 40 m                                        |
| Pompa doppia montata 80 m³/h – prevalenza 25 m<br>Twin pump, factory - fitted, 80 m³/h – head 25 m |
| Pompa doppia montata 80 m³/h – prevalenza 40 m<br>Twin pump, factory - fitted, 80 m³/h – head 40 m |

## Serbatoi e vasi di espansione / Tanks and expansion vessels

|                                                                                                                                                                         |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Serbatoio idraulico esterno 1000 l in AISI 304 con pompa doppia in AISI 304 montata<br>External water tank 1000 l, AISI 304 with factory - fitted twin AISI 304 pump    |
| Serbatoio idraulico esterno 1500 l in AISI 304 con pompa doppia in AISI 304 montata<br>External water tank 1500 l, AISI 304 with factory - fitted twin AISI 304 pump    |
| Serbatoio idraulico esterno 2000 l in AISI 304 con pompa doppia in AISI 304 montata<br>External water tank 2000 l, AISI 304 with factory - fitted twin AISI 304 pump    |
| Serbatoio idraulico esterno 3000 l in AISI 304 con pompa doppia in AISI 304 montata<br>External water tank 3000 l, AISI 304 with factory - fitted twin AISI 304 pump    |
| Serbatoio idraulico esterno 5000 l in AISI 304 con pompa doppia in AISI 304 montata<br>External water tank 5000 l, AISI 304 with factory - fitted twin AISI 304 pump    |
| Serbatoio idraulico esterno 1000 l in AISI 304 con pompa doppia montata<br>External water tank 1000 l, AISI 304 with factory - fitted twin pump                         |
| Serbatoio idraulico esterno 1500 l in AISI 304 con pompa doppia montata<br>External water tank 1500 l, AISI 304 with factory - fitted twin pump                         |
| Serbatoio idraulico esterno 2000 l in AISI 304 con pompa doppia montata<br>External water tank 2000 l, AISI 304 with factory - fitted twin pump                         |
| Serbatoio idraulico esterno 3000 l in AISI 304 con pompa doppia montata<br>External water tank 3000 l, AISI 304 with factory - fitted twin pump                         |
| Serbatoio idraulico esterno 5000 l in AISI 304 con pompa doppia montata<br>External water tank 5000 l, AISI 304 with factory - fitted twin pump                         |
| Serbatoio idraulico esterno 1000 l in AISI 304 con pompa singola in AISI 304 montata<br>External water tank 1000 l, AISI 304 with factory - fitted single AISI 304 pump |
| Serbatoio idraulico esterno 1500 l in AISI 304 con pompa singola in AISI 304 montata<br>External water tank 1500 l, AISI 304 with factory - fitted single AISI 304 pump |
| Serbatoio idraulico esterno 2000 l in AISI 304 con pompa singola in AISI 304 montata<br>External water tank 2000 l, AISI 304 with factory - fitted single AISI 304 pump |
| Serbatoio idraulico esterno 3000 l in AISI 304 con pompa singola in AISI 304 montata<br>External water tank 3000 l, AISI 304 with factory - fitted single AISI 304 pump |

|                                                                                                                                                                                              |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Serbatoio idraulico esterno 5000 l in AISI 304 con pompa singola in AISI 304 montata<br>External water tank 5000 l, AISI 304 with factory - fitted single AISI 304 pump                      |
| Serbatoio idraulico esterno 1000 l in AISI 304 con pompa singola montata<br>External water tank 1000 l, AISI 304 with factory - fitted single pump                                           |
| Serbatoio idraulico esterno 1500 l in AISI 304 con pompa singola montata<br>External water tank 1500 l, AISI 304 with factory - fitted single pump                                           |
| Serbatoio idraulico esterno 2000 l in AISI 304 con pompa singola montata<br>External water tank 2000 l, AISI 304 with factory - fitted single pump                                           |
| Serbatoio idraulico esterno 3000 l in AISI 304 con pompa singola montata<br>External water tank 3000 l, AISI 304 with factory - fitted single pump                                           |
| Serbatoio idraulico esterno 5000 l in AISI 304 con pompa singola montata<br>External water tank 5000 l, AISI 304 with factory - fitted single pump                                           |
| Serbatoio idraulico esterno 1000 l pressurizzato in acciaio zincato con pompa doppia montata<br>External water tank 1000 l, pressurised, galvanised steel with factory - fitted twin pump    |
| Serbatoio idraulico esterno 1500 l pressurizzato in acciaio zincato con pompa doppia montata<br>External water tank 1500 l, pressurised, galvanised steel with factory - fitted twin pump    |
| Serbatoio idraulico esterno 2000 l pressurizzato in acciaio zincato con pompa doppia montata<br>External water tank 2000 l, pressurised, galvanised steel with factory - fitted twin pump    |
| Serbatoio idraulico esterno 3000 l pressurizzato in acciaio zincato con pompa doppia montata<br>External water tank 3000 l, pressurised, galvanised steel with factory - fitted twin pump    |
| Serbatoio idraulico esterno 5000 l pressurizzato in acciaio zincato con pompa doppia montata<br>External water tank 5000 l, pressurised, galvanised steel with factory - fitted twin pump    |
| Serbatoio idraulico esterno 1000 l pressurizzato in acciaio zincato con pompa singola montata<br>External water tank 1000 l, pressurised, galvanised steel with factory - fitted single pump |
| Serbatoio idraulico esterno 1500 l pressurizzato in acciaio zincato con pompa singola montata<br>External water tank 1500 l, pressurised, galvanised steel with factory - fitted single pump |
| Serbatoio idraulico esterno 3000 l pressurizzato in acciaio zincato con pompa singola montata<br>External water tank 3000 l, pressurised, galvanised steel with factory - fitted single pump |

# Optional — 2/3 Options

PROS6-A670-OSB · 75 optional disponibili per questa configurazione · 75 options available for this configuration

## Serbatoi e vasi di espansione / Tanks and expansion vessels

Serbatoio idraulico esterno 5000 l pressurizzato in acciaio zincato con pompa singola montata

External water tank 5000 l, pressurised, galvanised steel with factory-fitted single pump

Vaso di espansione 100 l

Expansion vessel 100 l

Vaso di espansione 200 l

Expansion vessel 200 l

Vaso di espansione 300 l

Expansion vessel 300 l

Vaso di espansione 500 l

Expansion vessel 500 l

Vaso di espansione 750 l

Expansion vessel 750 l

## Valvole e filtri / Valves and filters

Elettrovalvola 2"

Solenoid valve 2"

Elettrovalvola 3"

Solenoid valve 3"

Elettrovalvola 4"

Solenoid valve 4"

Elettrovalvola 5"

Solenoid valve 5"

Elettrovalvola 6"

Solenoid valve 6"

Filtro a dischi in acciaio 2"

Steel disc filter 2"

Filtro a dischi in acciaio 3"

Steel disc filter 3"

Filtro a dischi in acciaio 4"

Steel disc filter 4"

Filtro meccanico a Y 2"

Y-strainer 2"

Filtro meccanico a Y 3"

Y-strainer 3"

Filtro meccanico a Y 4"

Y-strainer 4"

Filtro meccanico a Y 5"

Y-strainer 5"

Filtro meccanico a Y 6"

Y-strainer 6"

Valvola a 3 vie 2" miscelatrice

3-way mixing valve 2"

Valvola a 3 vie 3" miscelatrice

3-way mixing valve 3"

Valvola a 3 vie 4" miscelatrice

3-way mixing valve 4"

Valvola a 3 vie 5" miscelatrice

3-way mixing valve 5"

Valvola a 3 vie 6" miscelatrice

3-way mixing valve 6"

Valvola a 3 vie 2" on/off

3-way valve 2" on/off

Valvola a 3 vie 3" on/off

3-way valve 3" on/off

Valvola a 3 vie 4" on/off

3-way valve 4" on/off

Valvola a 3 vie 5" on/off

3-way valve 5" on/off

Valvola a 3 vie 6" on/off

3-way valve 6" on/off

## Circuito frigorifero / Refrigerant circuit

Cofanatura insonorizzante compressori

Compressor acoustic enclosure

Compressore inverter

Inverter compressor

Desurriscaldatore a piastre

Plate desuperheater

Rilevatore fughe gas

Refrigerant leak detector

## Free cooling / Free cooling

Free cooling 500 kW – 12 ventilatori (rif. 8)

Free cooling 500 kW – 12 fans (ref. 8)

## Ventilazione e clima / Fans and climate

Inverter ventilatori

Fan inverter

Ventilatori EC

EC fans

## Controllo e comunicazione / Control and communication

Comando remoto con display (cavo escluso)

Remote control with display (cable not included)

Comunicazione RS485 Modbus RTU / TCP-IP (cavo escluso)

RS485 Modbus RTU / TCP-IP communication (cable not included)

Comunicazione RS485 verso BACnet (cavo escluso)

RS485 to BACnet communication (cable not included)

## Quadro elettrico / Electrical panel

Analizzatore di rete

Power analyser

# Optional — 3/3

## Options

PROS6-A670-OSB · 75 optional disponibili per questa configurazione · 75 options available for this configuration

### Quadro elettrico / Electrical panel

Kit invernale (inverter ventilatori + resistenza quadro + resistenza evaporatore + termostato) fino a -25 °C

Winter kit (fan inverter + panel heater + evaporator heater + thermostat) down to -25 °C

### Installazione / Installation

Antivibranti a molla

Spring anti-vibration mounts

Antivibranti in gomma

Rubber anti-vibration mounts

## Note Notes

- Prestazioni con acqua pura, salto termico 5 K. Le condizioni di riferimento sono indicate in ogni tabella.
- Con miscele di glicole la resa frigorifera si corregge con i coefficienti della tabella dedicata; potenza assorbita ed EER restano invariati.
- Dimensioni ricavate dai disegni quotati (tolleranza  $\pm 10$  mm): da confermare in fase d'ordine.
- Dati elettrici, carica di refrigerante, pesi e attacchi idraulici sono forniti su richiesta per la configurazione scelta.
- Procryon si riserva di modificare dati e caratteristiche senza preavviso, per il miglioramento continuo del prodotto.
- Performance with pure water, 5 K temperature difference. Reference conditions are stated in each table.
- With glycol mixtures, cooling capacity is corrected with the coefficients in the dedicated table; power input and EER are unchanged.
- Dimensions taken from dimensional drawings (tolerance  $\pm 10$  mm): to be confirmed at order stage.
- Electrical data, refrigerant charge, weights and water connections are available on request for the chosen configuration.
- Procryon reserves the right to change data and features without notice, as part of continuous product improvement.

## Parliamoci Let's talk

Progettazione, intervento urgente, valutazione di un impianto esistente, supporto tecnico e intervento rapido — 24/7.

System design, urgent response, assessment of an existing plant, technical support and rapid response — 24/7.

Procryon — Smart Cooling Engineering S.r.l.  
Via Ugo Vaglia 10 — Gavardo (BS) — Italia  
[www.procryon.com](http://www.procryon.com)



**PROCRYON**

Procryon — Smart Cooling Engineering S.r.l.  
[www.procryon.com](http://www.procryon.com)

Continuity by design